Qingzhao Zhang

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RESEARCH INTERESTS

System security (e.g., cyber-physical systems, autonomous driving), software security (e.g., program analysis, formal verification), AI security (e.g., adversarial attacks, robustness).

EDUCATION

University of Michigan — *Sep 2019 – May 2025*
Ph.D. *in Computer Science and Engineering*

University of Michigan — *Sep 2019 – Dec 2023*
M.S. *in Computer Science and Engineering*

Shanghai Jiao Tong University — *Sep 2015 – May 2019*
B.E. *in Computer Engineering*

WORK EXPERIENCE

Assistant Professor | University of Arizona, ECE — *Aug 2025 - Present*
- Lead *Secure Autonomous Systems* (SAS) research group working on computer security and systems. We advance the security, resilience, and trustworthiness of emerging autonomous systems (e.g., self-driving vehicles, drones, and robots) through cross-layer design and analysis.

Research Assistant/Research Fellow | University of Michigan, EECS — *May 2020 - Aug 2025*
Supervisor: Prof. Z. Morley Mao
- Research on software/system for cyber-physical system safety — *AVChecker*, the first traffic rule compliance checker on autonomous driving software; *SmtConf*, safety vetting of industrial control system configuration.
- Research on adversarial machine learning on cyber-physical systems — Adversarial attack and mitigation on trajectory prediction on autonomous driving; Analyzed data fabrication vulnerability on collaborative perception.
- Research on robustness of vehicular network — *RAO*, collaborative perception with asynchronous sensors.
- Research on robust perception algorithms of autonomous driving, and large language model security/efficiency.

Software Engineer Intern | Google — *May 2023 - July 2023*
- Designed and implemented formal verification solutions to enhance the security properties of an embedded system kernel (based on open-sourced Tock OS), involving modular model checking and theorem proving.

Software Engineer Intern | Google — *May 2022 - July 2022*
- Designed and implemented static analysis checks based on Android Lint for Google's Android tests, which was deployed to assist Google developers to write high-quality unit tests.

SELECTED PUBLICATIONS

Conference Papers

- **[ACNS '26]** Anrin Chakraborti, **Qingzhao Zhang**, Jingjia Peng, Z. Morley Mao, Michael K. Reiter. "Automatic Teller Machines for Offline Ecash", *The 24th International Conference on Applied Cryptography and Network Security*.
- **[S&P '26]** Jiarui Li, Joey Brewington, **Qingzhao Zhang**, Z. Morley Mao. "Banshee: Target Switch Attacks on Gimbal-Stabilized Visual Tracking Systems via Acoustic Injection", *The 47th IEEE Symposium on Security and Privacy*.
- **[COLM '25]** Shuowei Jin, Yongji Wu, Haizhong Zheng, **Qingzhao Zhang**, Matthew Lentz, Z Morley Mao, Atul Prakash, Feng Qian, Danyang Zhuo. "Plato: Plan to Efficiently Decode for Large Language Model Inference", *Conference on Language Modeling*.
- **[ICML '25]** Shuowei Jin*, Xueshen Liu*, **Qingzhao Zhang**, Z. Morley Mao. "Compute Or Load KV Cache? Why Not Both?", *Forty-Second International Conference on Machine Learning*.
- **[ICLR '25]** Minkyoung Cho, Yulong Cao, Jiachen Sun, **Qingzhao Zhang**, Marco Pavone, Jeong Joon Park, Heng Yang, Z. Morley Mao. "Cocoon: Robust Multi-Modal Perception with Uncertainty-Aware Sensor Fusion", *The International Conference on Learning Representations*.

- **[USENIX Security '24] Qingzhao Zhang**, Shuowei Jin, Ruiyang Zhu, Jiachen Sun, Xumiao Zhang, Qi Alfred Chen, Z. Morley Mao. "On Data Fabrication in Collaborative Vehicular Perception: Attacks and Countermeasures", *Proceedings of the 33nd USENIX Security Symposium*.
- **[ICLR '24]** Jiachen Sun, Haizhong Zheng, **Qingzhao Zhang**, Atul Prakash, Z. Morley Mao, Chaowei Xiao. "CALICO: Self-Supervised Camera-LiDAR Contrastive Pre-training for BEV Perception", *The International Conference on Learning Representations*.
- **[MobiCom '23] Qingzhao Zhang***, Xumiao Zhang*, Ruiyang Zhu*, Fan Bai, Mohammad Naserian, Z. Morley Mao. "Robust Real-time Multi-vehicle Collaboration on Asynchronous Sensors", *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*.
- **[RAID '22] Qingzhao Zhang**, Xiao Zhu, Mu Zhang, Z. Morley Mao. "Automated Runtime Mitigation for Misconfiguration Vulnerabilities in Industrial Control Systems", *Proceedings of the 25th International Symposium on Research in Attacks, Intrusions and Defenses*.
- **[CVPR '22] Qingzhao Zhang**, Shengtuo Hu, Jiachen Sun, Qi Alfred Chen, Z. Morley Mao. "On adversarial robustness of trajectory prediction for autonomous vehicles", *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- **[IV '22]** Ze Zhang, Sanjay Sri Vallabh Singapuram, **Qingzhao Zhang**, David Ke Hong, Brandon Nguyen, Z. Morley Mao, Scott Mahlke, Qi Alfred Chen. "AVMaestro: A Centralized Policy Enforcement Framework for Safe Autonomous-driving Environments", *IEEE Intelligent Vehicles Symposium*.
- **[AsiaCCS '22]** Shengtuo Hu, **Qingzhao Zhang**, André Weimerskirch, Z. Morley Mao. "GateKeeper: A Gateway-based Broadcast Authentication Protocol for the In-vehicle Ethernet", *Proceedings of the ACM on Asia Conference on Computer and Communications Security*.
- **[SIGMETRICS '21] Qingzhao Zhang**, David Ke Hong, Ze Zhang, Qi Alfred Chen, Scott Mahlke, Z Morley Mao. "A systematic framework to identify violations of scenario-dependent driving rules in autonomous vehicle software", *Proceedings of the ACM on Measurement and Analysis of Computing Systems*.
- **[SANER '20] Qingzhao Zhang***, Yizhuo Wang*, Juanru Li, Siqi Ma. "Ethploit: From fuzzing to efficient exploit generation against smart contracts", *IEEE 27th International Conference on Software Analysis, Evolution and Reengineering*.
- **[FC '20]** Tsz Hon Yuen, Shi-feng Sun, Joseph K Liu, Man Ho Au, Muhammed F Esgin, **Qingzhao Zhang**, Dawu Gu. "Ringct 3.0 for blockchain confidential transaction: Shorter size and stronger security", *Financial Cryptography and Data Security: 24th International Conference*.

Journal Papers

- **[TDSC]** Yanxue Jia, Shi-Feng Sun, Yuncong Zhang, **Qingzhao Zhang**, Ning Ding, Zhiqiang Liu, Joseph K Liu, Dawu Gu. "PBT: A New Privacy-Preserving Payment Protocol for Blockchain Transactions", *IEEE Transactions on Dependable and Secure Computing*.

Workshop and Short Papers

- **[LAMPS '25] Qingzhao Zhang**, Ziyang Xiong, Z. Morley Mao. "Safeguard is a Double-edged Sword: Denial-of-service Attack on Large Language Models", *2nd ACM Workshop on Large AI Systems and Models with Privacy and Safey Analysis*.
- **[VehicleSec '25]** Jiarui Li, Joseph Brewington, **Qingzhao Zhang**, Z. Morley Mao. "WIP: Hijacking Attacks on UAV Follow-Me Systems in Realistic Scenarios", *3rd USENIX Symposium on Vehicle Security and Privacy*.
- **[CPSIoTSec '24] Qingzhao Zhang**, Z. Morley Mao. "Stealthy Data Fabrication in Collaborative Vehicular Perception", *the 6th Workshop on CPS and IoT Security*.
- **[SRML '22]** Jiachen Sun, **Qingzhao Zhang**, Bhavya Kailkhura, Zhiding Yu, Chaowei Xiao, Z. Morley Mao. "ModelNet40-C: a robustness benchmark for 3D point cloud recognition under corruption", *Workshop on Socially Responsible Machine Learning*.
- **[FEAST '20]** Ze Zhang, **Qingzhao Zhang**, Brandon Nguyen, Sanjay Sri Vallabh Singapuram, Z Morley Mao, Scott Mahlke. "Automatic Feature Isolation in Network Protocol Software Implementations", *Proceedings of the 2020 ACM Workshop on Forming an Ecosystem Around Software Transformation*.

Preprints

- Ruiyang Zhu, Yuehan He, Boyuan Zheng, Zesen Zhao, Ahmad Chalhoub, **Qingzhao Zhang**, Z. Morley Mao. "CLAP: Contrastive Latent-space Prompt Optimization for End-to-end Autonomous Driving", *arXiv.org e-Print archive*.
- Yutong Liu, Chenyi Wang, Ming F. Li, **Qingzhao Zhang**. "Adversarial Trust Poisoning in Vehicular Collaborative Perception", *arXiv.org e-Print archive*.
- **Qingzhao Zhang**, Runting Zhang, Z. Morley Mao. "From Stealthy Data Fabrication to Unsafe Driving: Realistic Scenario Attacks on Collaborative Perception", *arXiv.org e-Print archive*.
- Shuo Ju, **Qingzhao Zhang**, Huashan Chen, Xuheng Wang, Haotang Li, Wanqian Zhang, Feng Liu, Kebin Peng, Sen He. "Still Camouflage, Moving Illusion: View-Induced Trajectory Manipulation in Autonomous Driving", *arXiv.org e-Print archive*.
- **Qingzhao Zhang**, Shaocheng Luo, Z Morley Mao, Miroslav Pajic, Michael K Reiter. "SoK: How Sensor Attacks Disrupt Autonomous Vehicles: An End-to-end Analysis, Challenges, and Missed Threats", *arXiv.org e-Print archive*.
- Fangzhou Wu, **Qingzhao Zhang**, Ati Priya Bajaj, Tiffany Bao, Ning Zhang, Ruoyu Wang, Chaowei Xiao. "Exploring the Limits of ChatGPT in Software Security Applications", *arXiv.org e-Print archive*.
- R. Spencer Hallyburton, **Qingzhao Zhang**, Z. Morley Mao, Miroslav Pajic. "Partial-Information, Longitudinal Cyber Attacks on LiDAR in Autonomous Vehicles", *arXiv.org e-Print archive*.
- Jiachen Sun, **Qingzhao Zhang**, Bhavya Kailkhura, Zhiding Yu, Chaowei Xiao, Z. Morley Mao. "Benchmarking Robustness of 3D Point Cloud Recognition Against Common Corruptions", *arXiv.org e-Print archive*.

TEACHING AND MENTORSHIP

Course Instructor, University of Arizona
- Fall 2026, ECE 509 Cyber Security - Concept, Theory, Practice.
- Spring 2026, ECE 402C/502C Operating System Design.

PhD Students, University of Arizona
- Yutong Liu, started in Fall 2025.
- Yameng Ge, started in Fall 2026.
- Xianwu Zhao, started in Fall 2026.

Undergraduate Students, University of Arizona
- Ahmed Mokhtar Abakar, UROC.

Supervisor of Undergraduate Research, University of Michigan *2022*
- Multidisciplinary Design Program (MDP) at University of Michigan. Lead cybersecurity research and supervised a team of four students.

Research Mentorship, University of Michigan *2020-2024*
- Research on program analysis, autonomous driving, adversarial machine learning.
- Mentees: Charles Ziegenbein Jr., Kevin Zhang, Andrew Wei, Xingyu Wang, Jingjia Peng, Ziyang Xiong, Runting Zhang.

SERVICES

- Conference Committee Member: NDSS 2027, S&P 2027, AsiaCCS 2027, CCS 2026/2025
- Conference Reviewer: NeurIPS 2026, COLM 2026, CVPR 2026, ECCV 2026, ICLR 2026/2025, ICRA 2025, ACM MM 2024, IV 2024/2023
- Journal Reviewer: TDSC, IoT-J, Sensors, TIFS, ITSM
- Workshop Committee Member: RICSS 2024, RICSS 2025
- Artifact Reviewer: Usenix Security 2022
- Pre-review Task Force: NSDI 2025

TALKS

- Enhancing Security, Safety, and Reliability of Cyber-physical Systems

Presentation at Midwest Security Workshop *11/16/2024*

- Stealthy Data Fabrication in Collaborative Vehicular Perception

Presentation at 6th CPSIoTSec Workshop, co-located with CCS 2024 *10/18/2024*
Poster at Midwest Security Workshop *11/16/2024*

- On data fabrication in collaborative vehicular perception: attacks and countermeasures

Poster at Athena institute (NSF AI Institute) annual showcase (best poster award) *5/3/2023*
Presentation at USENIX Security 2024 *8/16/2024*

- Automated runtime mitigation for timing-based safety hazards in industrial controllers

Virtual presentation at RAID 2022 *10/27/2022*

- On adversarial robustness of trajectory prediction for autonomous vehicles

Poster at Athena institute (NSF AI Institute) annual showcase *8/18/2022*
Poster at CVPR 2022 *6/23/2022*

- Robustness of applications for autonomous vehicles

Presentation at Workshop on Future Automotive Research Datasets *11/19/2021*

- A systematic framework for checking driving rule compliance in autonomous vehicle software

Virtual presentation at SIGMETRICS 2021 *6/16/2021*

HONORS AND AWARDS

- Student Travel Grant, Usenix Security 2024, SIGMETRICS 2021, CCS 2021 *2019 – 2024*
- Rackham Student Fellowship, University of Michigan *2019*
- Academic Excellence Scholarship of SJTU (top 5%) *2016, 2017, 2018*